

RESEARCH ARTICLE



Structure and cytoarchitecture of the retrosplenial cortex in the Cairo spiny mouse (*Acomys cahirinus*)

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Abstract

The retrosplenial cortex is a component of the limbic cortex involved in visually guided memory tasks, including navigation. Using neurochemical markers for the general neuronal population (NeuN), pyramidal neurons (SMI-32), and interneurons (calbindin 28 kDa), we examined the location and cytoarchitectonic divisions of the retrosplenial cortex in the Cairo spiny mouse (*Acomys cahirinus*). Two distinct regions corresponding to the granular and dysgranular areas (areas 29 and 30) were identified. The criterion for defining the granular cortex was the presence of dendritic bundles located within layers I and II, exhibiting intense SMI-32 staining. SMI-32-positive neuronal somas were predominantly found in layers II and V of area 29 and in layers II–V of area 30. In both areas, calbindin-positive neuronal somas were mainly located in layer II. The total neuronal population was greater in area 29, whereas both SMI-32- and calbindin-positive populations were more numerous in area 30. Based on these findings, we propose that some fundamental features of the retrosplenial cortex in the *Acomys* are quite similar to those observed in laboratory rodents. At the same time, the retrosplenial cortex in the *acomys* is larger than in the mouse brain.

KEYWORDS

Acomys cahirinus, area 29, area 30, calbindin, NeuN, retrosplenial cortex, SMI-32

1 | INTRODUCTION

The retrosplenial cortex is part of the limbic cortex, located immediately behind the splenium, the most caudal part of the corpus callosum (Vann et al., 2009). In rodents, it is situated at the transition zone between the lateral and medial surfaces of the cerebral hemispheres and represents one of the largest cortical regions (Aggleton et al., 2021). In primates, the retrosplenial cortex is found deep within the medial brain surface (Brodman, 1909; Kobayashi & Amaral, 2000). In both groups, it encompasses granular and dysgranular areas, which are Brodmann's areas 29 and 30, respectively

(Aggleton et al., 2021). Some features of the retrosplenial cortex are similar between rodents and primates (Alexander et al., 2023; Vann et al., 2009).

The retrosplenial cortex is considered a “nodal point” for information transfer between multiple brain regions, including the hippocampal formation, neocortex, basal forebrain, and thalamus (Kobayashi & Amaral, 2003; Kondo & Zaborszky, 2025; Robertson & Kaitz, 1981; Swanson & Cowan, 1977; Wyss & van Groen, 1992b) and it plays a key role in learning and memory (Wyss & van Groen, 1992b; see Alexander et al., 2023). In humans, activation of the retrosplenial cortex has been observed during autobiographical memory tasks, navigation, and

prospective thinking (see Vann et al., 2009). Its role in integrating sensory and motor systems with integrative regions such as the anterior thalamus and subiculum is believed to be unique (Alexander et al., 2023). One of the primary functions of the retrosplenial cortex is supporting visually guided memory behaviors, including navigation (Alexander & Nitz, 2015; Murray et al., 1989; Nelson et al., 2015). Research indicates that the granular retrosplenial cortex primarily receives hippocampal and parahippocampal information, while the dysgranular retrosplenial cortex primarily processes visual information (Aggleton et al., 2021).

In the present study, we investigated the retrosplenial cortex in the Cairo spiny mouse (*Acomys cahirinus*, referred to as the *acomys*). This species was recently introduced in neuroscience as an animal model with some unique features, including highly developed regenerative abilities (Kidd et al., 2024; Maden & Brant, 2019; Nogueira-Rodrigues et al., 2022) and menstrual, not estral cycle (Bellofiore et al., 2017; McKenna et al., 2020). *Acomys* is also a precocial rodent similar to degu (Long & Ebensperger, 2010) and Guinea pig (Kalusa et al., 2021) and has a tendency to develop type 2 diabetes (Shafir et al., 2006). Note that *Acomys* species are characterized by an extended longevity compared to laboratory mice: 4 years versus 1–2 years (Bartkowska et al., 2022; Bellofiore et al., 2021; Haughton et al., 2016).

We employed three neurochemical markers: (1) NeuN, a general neuronal marker (Mullen et al., 1992), (2) non-phosphorylated heavy-chain neurofilaments, which serve as markers for large neurons involved in long-lasting projections; these include pyramidal neurons within the cortex, labeled by SMI-32 antibodies (Campbell & Morrison, 1989; Fuentes-Santamaria et al., 2006), and (3) calbindin (CB), a Ca^{2+} -binding protein used as a marker for cortical interneurons (DeFelipe et al., 1999).

2 | MATERIALS AND METHODS

2.1 | Animals

Care and experimental procedures were conducted in accordance with the European Parliament Directive 2010/63/EU on the protection of animals used in scientific research, as well as the guidelines of the National Institutes of Health Guide for the Care and Use of Laboratory Animals. The study was approved by the Ethics Commission of the Pavlov Institute of Physiology (Protocol #02/17; 17 February 2022). All animals were bred in the laboratory of neuromorphology at the Pavlov Institute of Physiology, Russian Academy of Sciences, with rearing conditions previously described (Shkorbatova et al., 2024). A total of five adult *A. cahirinus*

(both sexes: two females and three males), aged 15 months, were used in this study. Note that there were conflicting hypotheses on the phylogenetic classification of *A. cahirinus* and the similar *Acomys dimidiatus* species. The last one was assigned to the *A. cahirinus*–*dimidiatus* complex (Denys et al., 1992; Volobouev et al., 2007).

2.2 | Immunohistochemistry

All animals were deeply anesthetized with a mixture of Zoletil (Virbac, Carros, France; 20 mg/kg) and Xyla (Interchemie werken “De Adelaar” BV, Venray, Netherlands; 2 mg/kg, i.p.), followed by transcardial perfusion with 0.9% NaCl and 4% paraformaldehyde (Sigma Aldrich, St. Louis, USA). After perfusion, the brains were carefully removed from the skull and sequentially immersed in 20% and 30% sucrose. The brains were then cut into 40 μm frontal slices using a freezing microtome (Reichert, Vienna, Austria). For all animals, three slices at the antero-posterior (AP) level from -1.0 to -3.0 containing both hemispheres were analyzed for each antigen.

The immunohistochemical staining procedures were previously described (Merkulyeva et al., 2024). In brief, endogenous peroxidase activity was blocked with 0.3% H_2O_2 , and nonspecific immunoreactivity was inhibited by incubation in 3% bovine serum albumin (Biolot, St. Petersburg, Russia). Slices were incubated with primary monoclonal mouse antibodies against non-phosphorylated domains of high-weight neurofilaments (SMI-32 antibodies; BioLegend, San Diego, USA; Cat# 801701; RRID: AB_2564642; dilution 1:15,000), monoclonal mouse antibodies against NeuN (Millipore, Newark, USA; Cat# MAB377; RRID: AB_2298772; dilution 1:3000), and monoclonal mouse antibodies against CB 28 kDa (Sigma-Aldrich, St. Louis, USA; Cat# C9848; RRID: AB_476894; dilution 1:10,000) for 70 h. Following incubation, slices were treated with secondary biotinylated horse anti-mouse IgG antibodies (Vector Laboratories, Inc., USA; Cat# BA-2000; RRID: AB_2313581; dilution 1:600) for 24 h. The slices were then processed using the avidin-biotin complex (ABC Elite system, Vector Laboratories, Inc., USA) and diaminobenzidine solution (DAB; Sigma Aldrich, St. Louis, USA). Finally, the slices were washed, mounted, dehydrated, cleared, and placed under coverslips using Bio-Mount HM (Bio-Optica, Milan, Italy).

2.3 | Image analysis

Images were acquired using a computer setup equipped with an Olympus CX31 light microscope (Olympus

Corporation, Tokyo, Japan; 10× objective), the VideoZavr Scan 2.4 software package (VideoZavr, St. Petersburg, Russia), and a VideoZavr Standart VZ-18C23-B camera (VideoZavr, St. Petersburg, Russia). All image processing and measurements, specifically cell counts and neuronal soma areas across areas 29 and 30 of the retrosplenial cortex, were performed using the free Fiji ImageJ software (v. 2.14.0/1.54f) (Schindelin et al., 2012). All cell counts were done manually. Initially, clear criteria were established for identifying labeled cells. The first operator then marked each immunopositive cell with a dot on images of the slices, using the microscope to verify doubtful cases. Subsequently, a second operator reviewed the marked images and verified all markings. NeuN-labeling (NeuN+) was analyzed using two slices per animal, SMI-32 (SMI-32+) labeling—using six to eight slices, and CB (CB+) labeling—using six slices. The methods for defining and subdividing the retrosplenial cortex are described below. Note that cell counting at the SMI-32-labeled slices was complicated by the dense, darkly stained neuropil, which obscured somatic labeling.

2.4 | Statistics

Data are presented as mean $\pm$ standard deviation (SD). Normality was assessed using the Shapiro–Wilk test. For the paired inter-areal and intra-areal comparisons, a one-way Analysis of Variance (ANOVA) test with Bonferroni's multiple comparisons correction was employed. To compare overall cellular density and the area of the retrosplenial regions, the paired *t*-test was used. Despite the limited sample size in this study, the inter-individual variability for all parameters was small.

3 | RESULTS

3.1 | General location and division of the retrosplenial cortex

A general issue in studying the brain structures in any new animal model is an impossibility to directly translate the existing data from the well-known animal species. The retrosplenial cortex is located in the posterior wall region of the interhemispheric cleft and, in rodents, is comprised of laterally located dysgranular region and medially located granular region (van Groen & Wyss, 1990, 1992; Wyss & van Groen, 1992b). The boundary between these areas is characterized by distinct Nissl cytoarchitectonic features: the dysgranular area has wider layers II, III, and IV, and the granular area has more darkly stained and densely packed neurons of

layer II (van Groen & Wyss, 1992). Layers III–IV of the dysgranular area have a characteristic cuneiform shape being a well-defined feature of this cortex (Figure 1c–f). One of the prominent features of the granular retrosplenial cortex of rats is regular dendritic bundles formed in layer I by neurons of layers II and III (Ichinohe et al., 2003; Wyss et al., 1990). Using the SMI-32 antibody, similar regular dendritic bundles can be seen in the retrosplenial cortex of the *acomys* (red arrows on Figure 1d); they span the rostrocaudal AP levels from 1.1 to -4.3 mm from Bregma level where the caudal most part is correspondent to the superior colliculi level, and the rostral most part, to the striatum level (Figure 1a).

Depicting the borders of the retrosplenial cortex in the *acomys* brain, we noted that they do not correspond to the borders of the retrosplenial cortex in the mouse brain (according to the Allen Mouse Brain Connectivity Atlas; Allen Institute for Brain Science, 2011). A rostrocaudal extent of the retrosplenial cortex in relation to the total rostrocaudal extent of the cortex is 49.5% in the *acomys* and 40% in the mouse. It is notable that the rostrocaudal extent of the neighboring hippocampus in the *acomys* is also larger compared to that in the mouse (47.5% and 36%, respectively). The location of the retrosplenial cortex and hippocampus in relation to the bregma (stereotaxic values) is also different in the two species: the posterior borders are comparable (*acomys*: $[-4.3]$ and $[-4.3]$, mouse: $[-4.5]$ and $[-4.1]$), whereas the anterior borders are extended rostrally in *acomys* ($+1.1$) and $+0.8$ versus (-1.1) and (-1.0) . As a result, in the *acomys* brain, both the retrosplenial cortex and the hippocampus are larger compared to those in the mouse brain (Figure 1b).

An area of the identified layers of the retrosplenial cortex was measured using images with NeuN staining. In both areas 29 and 30, data is significantly dependent upon the layer (area 29: $F[1.825, 7.300] = 888.0$; $p < 0.0001$; area 30: $F[1.640, 6.561] = 318.5$; $p < 0.0001$). In area 29, layer V has the highest area, other layers have a three/four-fold lower area. In area 30, layer V has the highest area, layers III–IV and VI have comparable area values; layers I and II have a four-fold lower area. According to the post hoc analysis, in area 29, significant differences are observed between layer I and layers II, V ($p = 0.0007$, $p < 0.0001$), between layer II and layers III–VI ($p = 0.0005$, $p < 0.0001$, $p = 0.0002$), between layers III–IV and layers V, VI ($p < 0.0001$, $p = 0.0008$), and between layers V and VI ($p < 0.0001$). In area 30, significant differences are discovered between layer I and layers III–VI ($p = 0.0001$, $p < 0.0001$, $p = 0.0004$), layer II and layers III–VI ($p < 0.0001$, $p < 0.0001$, $p = 0.0009$), layers III–IV and layer V ($p = 0.0032$), and between layers V and VI ($p = 0.0044$). Inter-areal differences are

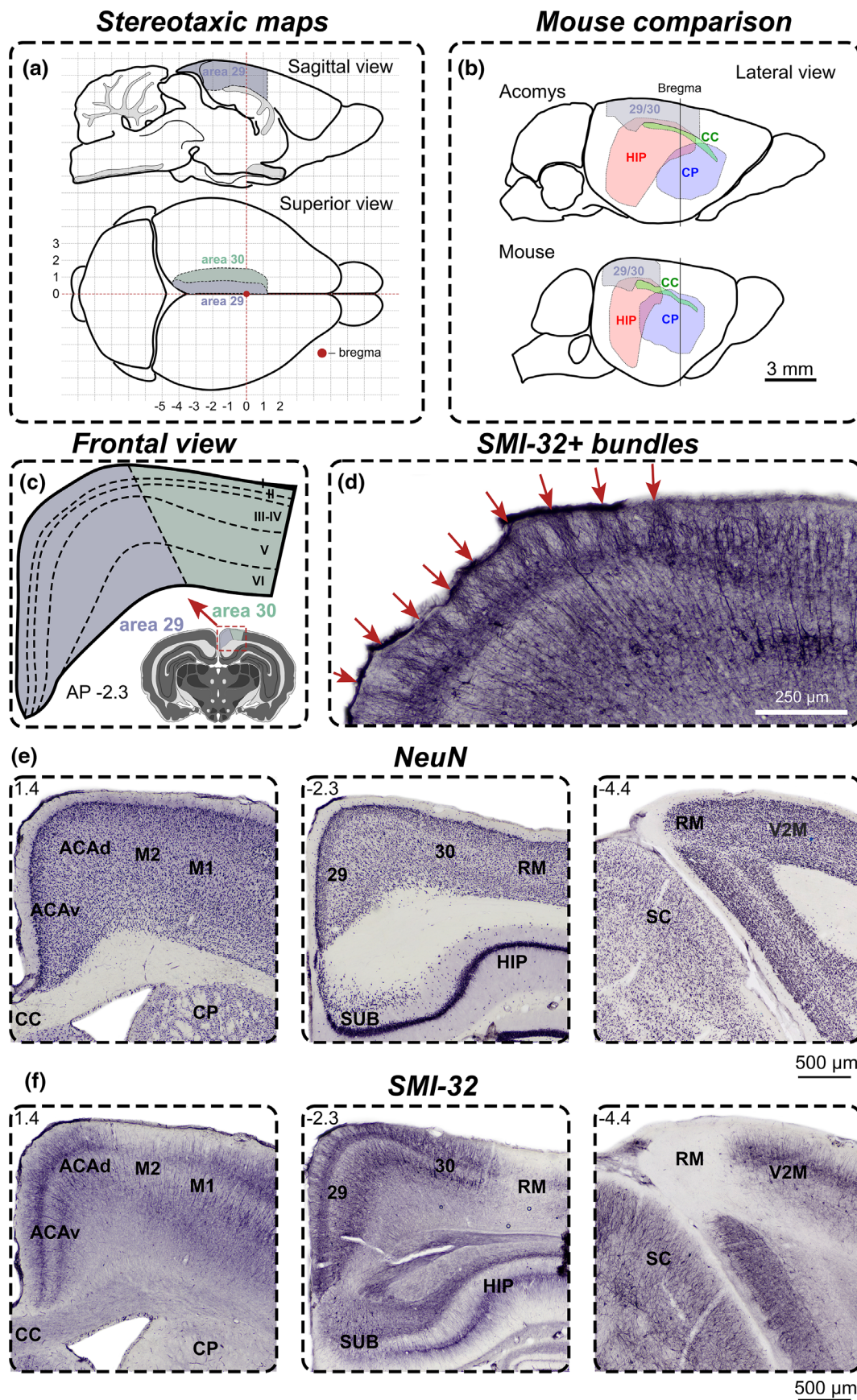


FIGURE 1 Legend on next page.

significant ($F[2.548, 10.19] = 632.8$; $p < 0.0001$); according to the post hoc analysis, significant differences are observed for layer I ($p = 0.0001$), layer V ($p < 0.0001$), and layer VI ($p = 0.0206$); in these cases, area 29 has larger layer's areas (Table 1). The total area of the granular area is higher compared to those in the dysgranular area, but this difference is insignificant (0.93 ± 0.06 vs. 0.59 ± 0.04 mm²; $p = 0.0001$).

3.2 | NeuN staining

Using NeuN staining, it is possible to clearly distinguish the borders between most layers (Figure 2a). The border between the lateral and medial regions of the retrosplenial cortex is marked by the width of the layers II and III: in the medial area (area 29) these layers compose a thin stripe, whereas in the lateral area (area 30), these layers expand from medial to lateral area borders, forming a cuneiform shape. In both areas, a hypocellular zone underlying layers II and III can be seen at some slices. This zone is more developed in area 29 and is correspondent to residual layer IV (Vogt & Peters, 1981). Because of the poor discernibility of this layer, layers III and IV are described together. In both retrosplenial areas, a well visible subgriseal layer (so-called "layer VII") can be seen. This layer composed of the persisted subplate neurons is obtained in species belonging to the *Rodentia*, *Insectivora*, *Paucituberculata*, *Paramelemorphia*, *Xenarthra*, and *Chiroptera* orders (Clancy & Cauller, 1999; Reep, 2000). The upper beginning of layer VII can be recognized by the vanishing of the vertically oriented cortical minicolumns. We did not analyze the NeuN+ neurons in this layer since we were not able to unambiguously distinguish the bottom border of layer VII, and the cellular density assessment cannot be conducted.

For both areas 29 and 30, the soma area of NeuN+ cells is significantly dependent upon the layer (area 29: $F[1.486, 5.943] = 70.38$; $p = 0.0001$, area 30: $F[2.255, 9.019] = 43.97$; $p < 0.0001$). The soma area of NeuN+ neurons in layers I of areas 29 and 30 is 38 ± 2 and

36 ± 4 μm², in layer II, is 31 ± 3 and 47 ± 5 μm², in layers III-IV, is 27 ± 2 and 43 ± 7 μm², in layer V, is 64 ± 8 and 72 ± 6 μm², and in layer VI, is 41 ± 3 and 42 ± 7 μm² (Figure 2b,c). For area 29, post hoc analysis revealed significant differences between layer I and layers III-IV, V ($p = 0.0084$, $p = 0.0049$), between layer II and layers V, VI ($p = 0.0068$, $p = 0.0161$), between layers III-IV and layers V, VI ($p = 0.0014$, $p = 0.0021$), and between layers V and VI ($p = 0.0048$). For area 30, post hoc analysis uncovered significant differences between layers I and V ($p = 0.0032$), between layers II and V ($p = 0.0091$), between layers III-IV and layer V ($p = 0.0002$), and between layers V and VI ($p = 0.0024$). Thereby, in area 29, the largest SMI-32+ cells are located in layers III-IV and V; however, in area 30, largest SMI-32+ cells are situated only in layer V. Inter-areal differences are significant ($F[2.404, 9.617] = 56.29$; $p < 0.0001$); according to the post hoc analysis, significant differences are observed for layers II ($p = 0.0052$), III-IV ($p = 0.0038$), and V ($p = 0.0113$): cells in area 30 are larger (Figure 2d).

Since two retrosplenial areas have different sizes, not an absolute number but a cellular density of NeuN+ neurons was analyzed. An overall cellular density is the same for areas 29 and 30 (4820 ± 240 cells/mm² vs. 4749 ± 556 cells/mm²; $p = 0.8370$). For both areas 29 and 30, the cellular density of NeuN+ cells is significantly dependent upon the layer ($F[1.291, 5.165] = 290.7$; $p < 0.0001$, $F[2.357, 9.429] = 128.6$; $p < 0.0001$). In area 29, the highest cellular density is observed in layers II and III-IV (Figure 5a); in area 30—in layers II, III-IV, and VI (Figure 5b). For area 29, post hoc analysis revealed significant differences between layer I and layers II, III-IV, V, and VI ($p = 0.0002$, $p < 0.0001$, $p < 0.0001$, $p < 0.0001$), layer II and layers III-IV, V, and VI ($p = 0.0120$, $p = 0.0005$, $p = 0.0007$), layers III-IV and layers V and VI ($p < 0.0001$, $p = 0.0003$), and between layers V and VI ($p = 0.0016$). For area 30, post hoc analysis demonstrated significant differences between layer I and layers II, III-IV, V, and VI ($p = 0.0006$, $p = 0.0003$, $p = 0.0016$, $p = 0.0031$), layer II and layers III-IV, V, and VI

FIGURE 1 Localization of retrosplenial areas 29 and 30 in the acornys brain. (a) Sagittal and superior views of the acornys brain with superimposed cortical areas 29 and 30; (b) Comparison of the rostrocaudal extent of areas 29 and 30 in the acornys and mouse. Dorsoventral and rostrocaudal borders of the shown brain structures are projected to the lateral view of the brains; (c) localization of areas 29 and 30 in a frontal brain slice; (d) SMI-32-immunopositive bundles in layer I of area 29. Dendritic bundles are marked with red arrows; (e) and (f) NeuN (e) and SMI-32 (f) labeling of cortical areas at three rostrocaudal levels: Areas 29 and 30 (middle columns, antero-posterior [AP] level -2.3), anterior cingulate areas (left columns, AP level $+1.4$), and visual areas (right columns, AP level -4.4); In (a), mediolateral and AP coordinates are shown on the right and bottom parts of the superior view, the red point is the bregma location. In (c), the AP level of the example slice is -2.3 . In (e) and (f), the AP level is shown in the upper left corner. 29—retrosplenial granular area, 30—retrosplenial dysgranular area, ACAd, dorsal anterior cingulate area; ACAv, ventral anterior cingulate area; CC, corpus callosum; CP, caudoputamen; HIP, hippocampus; M1, primary motor area; M2, secondary motor area; RM, retro-medial visual area; SUB, subiculum; V2M, medial secondary visual area. The scale bar is 3 mm (b), 500 μm (c, d) and 250 μm (e). The size of the square grid in (a) is 1×1 mm.

TABLE 1 Area of layers of the retrosplenial cortex (areas 29 and 30). Data (mm²) are presented as mean ± SD. Statistical comparisons were done using a two-way ANOVA test with Sidak multiple comparisons correction. Roman numerals—cortical layers.

	I	II	III-IV	V	VI
Area 29	0.15 ± 0.01	0.06 ± 0.01	0.14 ± 0.01	0.40 ± 0.01	0.18 ± 0.02
Area 30	0.06 ± 0.01; <i>p</i> = 0.0001; ***	0.05 ± 0.002; <i>p</i> = 0.1612; ns	0.14 ± 0.004; <i>p</i> = 0.9991; ns	0.19 ± 0.01; <i>p</i> < 0.0001; ****	0.14 ± 0.02; <i>p</i> = 0.0206; *

Note: * denotes inter-areal comparisons.
Abbreviation: ns, non-significant.

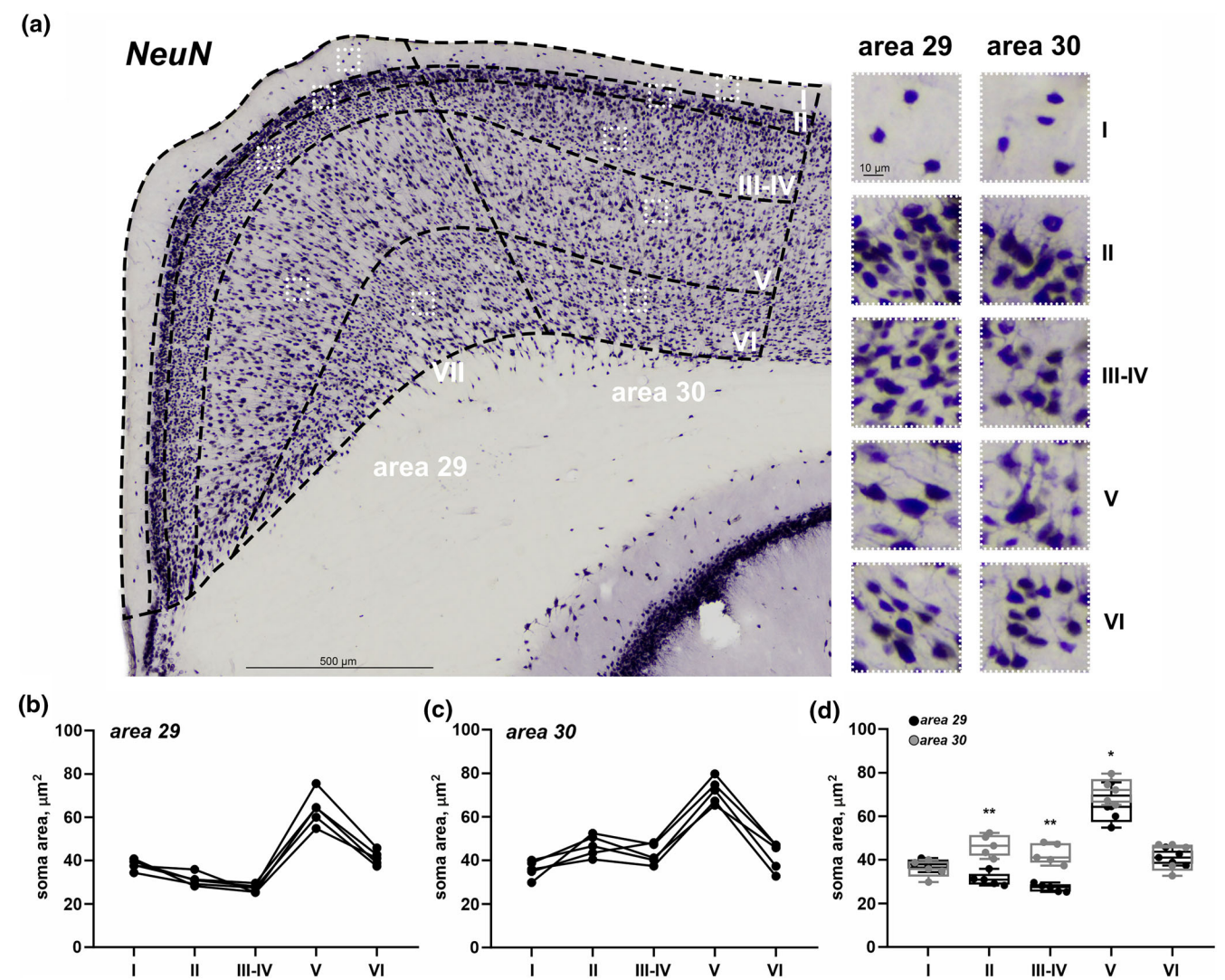


FIGURE 2 Distribution (a) and soma area (b–d) of NeuN-positive neurons in areas 29 and 30 of the cortex of the acomys. Roman numerals—cortical layers. Images within white dotted rectangles are enlarged in the right part of (a). (b and c) Individual data for areas 29 and 30, (d) averaged data. The mean values for animals are shown as individual points. **p* < 0.05, ***p* < 0.01. Post hoc Bonferroni's test.

(*p* = 0.0105, *p* = 0.0015, *p* = 0.0058), between layers III-IV and layer V (*p* = 0.0033), and between layers V and VI (*p* = 0.0398). Inter-areal differences are significant ($F[1.446, 5.785] = 153.0$; *p* < 0.0001); according to the post hoc analysis, significant differences are obtained for layers II (*p* = 0.0185) and III-IV (*p* = 0.0048): cellular density in area 29 is higher (Figure 5c).

3.3 | SMI-32 staining

Both areas 29 and 30 exhibit an intense SMI-32 staining. Laterally to this band, a region with the drastically weaker staining is located (Figure 3a); in mice, depending on the rostrocaudal level, this area corresponds to the secondary motor area, the posterior parietal association

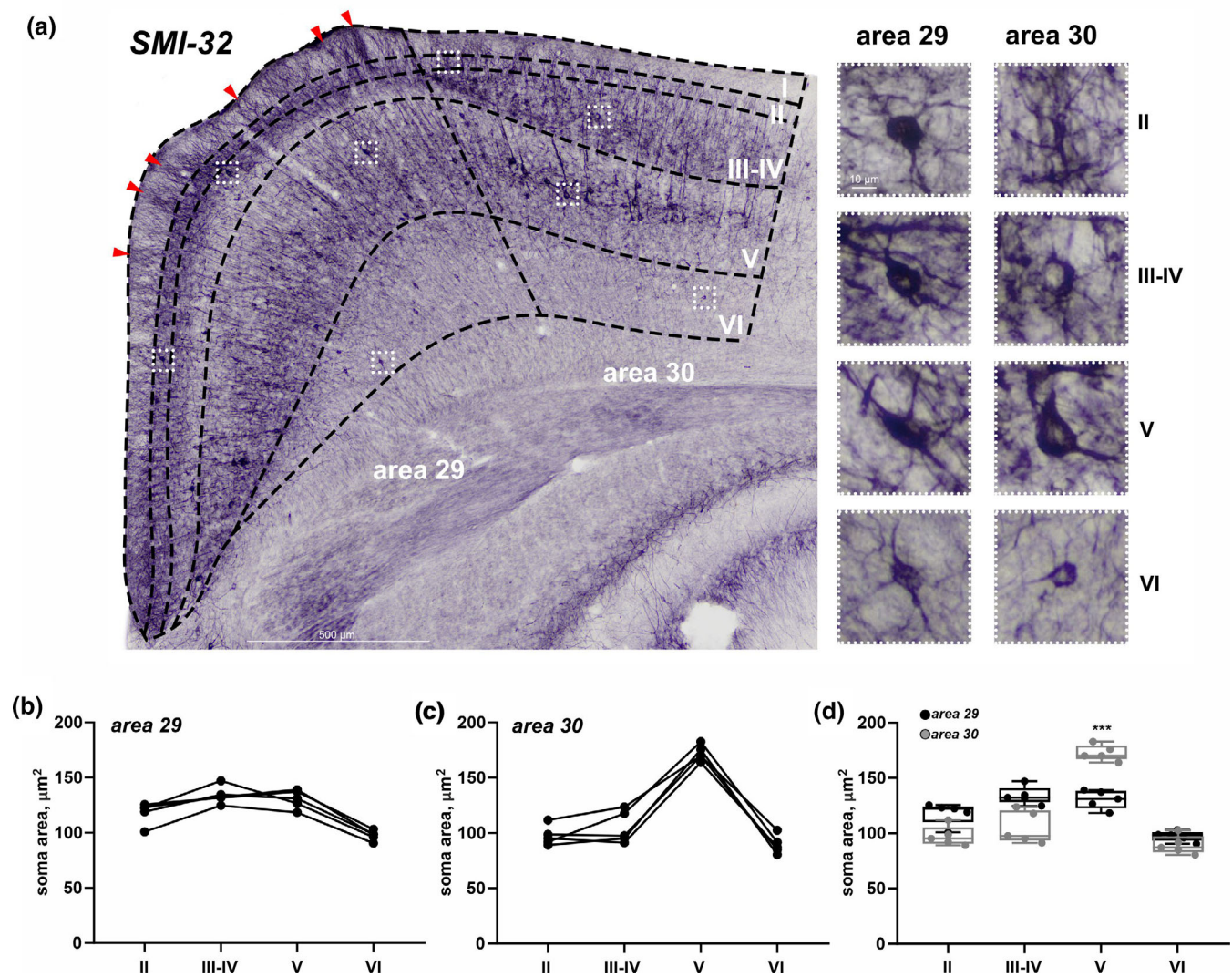


FIGURE 3 Distribution (a) and soma area (b–d) of SMI-32-positive neurons in areas 29 and 30 of the cortex of the acomys. Roman numerals—cortical layers. Images within white dotted rectangles are enlarged in the right part of (a). Dendritic bundles in (a) are marked with red arrows. (b and c) Individual data for areas 29 and 30, (d) averaged data. The mean values for animals are shown as individual points. *** $p < 0.001$. Post hoc Bonferroni's test.

areas or the extrastriate rostro-medial visual area (Van der Gucht et al., 2007). Medially, this band passes up to either the corpus callosum, postsubiculum, or subiculum, depending on the rostrocaudal level. An SMI-32-labeling consists of two bands containing both somatic and neuropil staining: layers II–III and layer V, a band containing mainly neuropil staining: layer I, and a band containing mainly somatic staining in layer VI (Figure 3a). The border between the two areas can be recognized using two criteria: (1) a replacement of layer V containing multiple large labeled neurons in area 30 by layer V containing few large labeled cells in area 29; (2) a clearly seen inclination of the cuneiform superficial layers in area 30 into area 29 (Figure 3a). Note that the former border is always situated more laterally compared to the inclination, even taking into account the offset of

neuronal microcolumns along the cortical curvature. We suppose this is a marker of the transition zone between the two areas.

Soma area of SMI-32+ cells was measured in all layers except for layer I due to the small sample size of cells in this layer. Soma area in both areas is significantly dependent upon the layer (area 29: $F[1.660, 6.641] = 46.37$; $p = 0.0002$, area 30: $F[2.034, 8.136] = 122.1$; $p < 0.0001$). In all layers, labeled cells are mainly pyramidal. The soma area of SMI-32+ cells in layer II is $118 \pm 10 \mu\text{m}^2$ in area 29 and $97 \pm 9 \mu\text{m}^2$ in area 30, in layers III–IV is 134 ± 8 and $105 \pm 15 \mu\text{m}^2$, in layer V is 131 ± 8 and $172 \pm 7 \mu\text{m}^2$, and in layer VI, is 97 ± 5 and $89 \pm 8 \mu\text{m}^2$ (Figure 3b,c). In area 29, post hoc analysis demonstrated significant differences between layer II and layers III–IV, V, and VI ($p = 0.0447$, $p = 0.0190$,

$p = 0.0061$), between layers III-IV and VI ($p = 0.0023$), and between layers V and VI ($p = 0.0004$). In area 30, post hoc analysis revealed significant differences between layer layers II and V ($p = 0.0008$), between layers III-IV and layers V and VI ($p = 0.0011$, $p = 0.0444$), and between layers V and VI ($p = 0.0003$). Thereby, in area 29, the largest SMI-32+ cells are situated in layers III-IV and V, in area 30, in layer V. Inter-areal differences are significant ($F[2.220, 8.880] = 73.15$; $p < 0.0001$); according to the post hoc analysis, significant differences are observed for layer V ($p = 0.0008$): cells in area 30 are larger (Figure 3d).

The general cellular density of SMI-32+ cells over the cortex is higher for the area 30 (147 ± 25 cells/mm² vs. 103 ± 12 cells/mm²; $p = 0.0054$). In both areas, the cellular density of SMI-32+ cells is significantly dependent upon the layer (area 29: $F[1.846, 7.384] = 32.62$; $p = 0.0002$, area 30: $F[1.586, 6.345] = 51.87$; $p = 0.0002$). In area 29, the highest cellular density is revealed in layers II and V (Figure 5d); in area 30—in layers II, III-IV, and V (Figure 5e). In area 29, post hoc analysis uncovered significant differences between layer I and layers II, III-IV, and V ($p = 0.0065$, $p = 0.0064$, $p = 0.0023$), between layers II and VI ($p = 0.0387$), between layers III-IV and layers V and VI ($p = 0.0083$, $p = 0.0450$), and between layers V and VI ($p = 0.0088$). In area 30, post hoc analysis revealed significant differences between layer I and layers II, III-IV, and V ($p = 0.0006$, $p = 0.0073$, $p = 0.0008$), between layer II and layers V and VI ($p = 0.0254$, $p = 0.0065$), between layers III-IV and layer VI ($p = 0.0237$), and between layers V and VI ($p = 0.0030$). Inter-areal differences are significant ($F[2.184, 8.735] = 41.89$; $p < 0.0001$); according to the post hoc analysis, significant difference is obtained for layer V ($p = 0.0015$); cellular density in area 30 is higher (Figure 5e).

Within area 29 and at the transition zone between areas 29 and 30, SMI-32 labeling in layers I and II is not homogeneous, and clearly visible patches of the labeled neuropil can be seen (Figure 1d). These patches were 89 ± 35 μm wide, varying from 25 to 215 μm . An averaged inter-patch distance was 170 ± 44 μm , varying from 91 to 320 μm .

3.4 | Calbindin staining

An intense CB-labeling is detected within area 30; however, area 29 has many fewer labeled neuronal somas and no neuropil labeling (Figure 4a), meaning that CB-labeling is a viable method for the retrosplenial cortex subdivision. In area 30, CB-labeling consists of a band containing neuropil staining corresponding to layers I–

IV, including somatic staining corresponding to layers II and IV, and a weaker developed band containing few labeled somas and weak neuropil staining corresponding to layers V and VI. In area 29, labeled cells can be mainly revealed in layers II and III. In both areas, but mostly in area 29, CB+ fibers are seen in layer VII.

Soma area of CB+ cells was measured in all layers except for layer I due to the small sample size of cells in this layer. Soma area of area 30 is significantly dependent upon the layer ($F[1.61, 5.443] = 28.94$; $p = 0.0017$) whereas in area 29, this link is insignificant ($F[1.157, 4.627] = 6.030$; $p = 0.0594$). In layer II of both areas, these cells are mainly round-shaped (Figure 4a); their soma area in area 29 is 51 ± 10 μm^2 and in area 30 is 50 ± 5 μm^2 (Figure 4b,c). In layers III-IV of both areas, solitary neurons having elongated or multiform soma can be seen (Figure 4a); their soma area is 57 ± 4 μm^2 in area 29 and 56 ± 5 μm^2 in area 30 (Figure 4b,c). In layers V-VI of area 30, and to a lesser extent, in area 29, oval-shaped or triangle-shaped cells with long dendrites are detected (Figure 4a); their soma area in area 29 is 62 ± 4 and 50 ± 5 μm^2 , for layers V and VI, and in area 30 is 44 ± 4 and 44 ± 4 μm^2 (Figure 4b,c). In area 29, post hoc analysis revealed significant differences between layers III-IV and layers V and VI ($p = 0.0432$, $p = 0.0350$), and between layers V and VI ($p = 0.0011$). In area 30, post hoc analysis uncovered significant differences between layers II and V ($p = 0.0114$), and layers III-IV and layers V, VI ($p = 0.0001$, $p = 0.0040$). In area 29, the largest CB+ cells are situated in layer V, in area 30, in layers III-IV (Figure 4b,c). Inter-areal differences are significant ($F[1.345, 5.380] = 13.05$; $p = 0.0111$); according to the post hoc analysis, significant differences are observed for layers V and VI ($p < 0.0001$, $p = 0.0010$): cells in area 29 are larger (Figure 4d).

The general cellular density of CB+ cells over the cortex is 10-fold higher for area 30 (584 ± 45 cells/mm² vs 46 ± 8 cells/mm²; $p = 0.0001$). In both areas, the cellular density of CB+ cells is significantly dependent upon the layer (area 29: $F[1.486, 5.946] = 52.25$; $p = 0.0002$, area 30: $F[1.292, 5.166] = 152.6$; $p < 0.0001$). In both areas, the highest cellular density is revealed in layer II (Figure 5g,h). In area 29, post hoc analysis demonstrated significant differences between layer I and layers II and III-IV ($p = 0.0065$, $p = 0.0292$), and between layer II and layers III-IV, V, and VI ($p = 0.0190$, $p = 0.0056$, $p = 0.0015$). In area 30, post hoc analysis revealed significant differences between layer I and layers II, III-IV ($p = 0.0005$, $p = 0.0022$), between layer II and layers III-IV, V, VI ($p = 0.0099$, $p = 0.0007$, $p = 0.0007$), and between layers III-IV and V, VI ($p = 0.0010$, $p = 0.0012$). Inter-areal differences are significant ($F[1.333, 5.333] = 175.6$; $p < 0.0001$); according to the post hoc analysis,

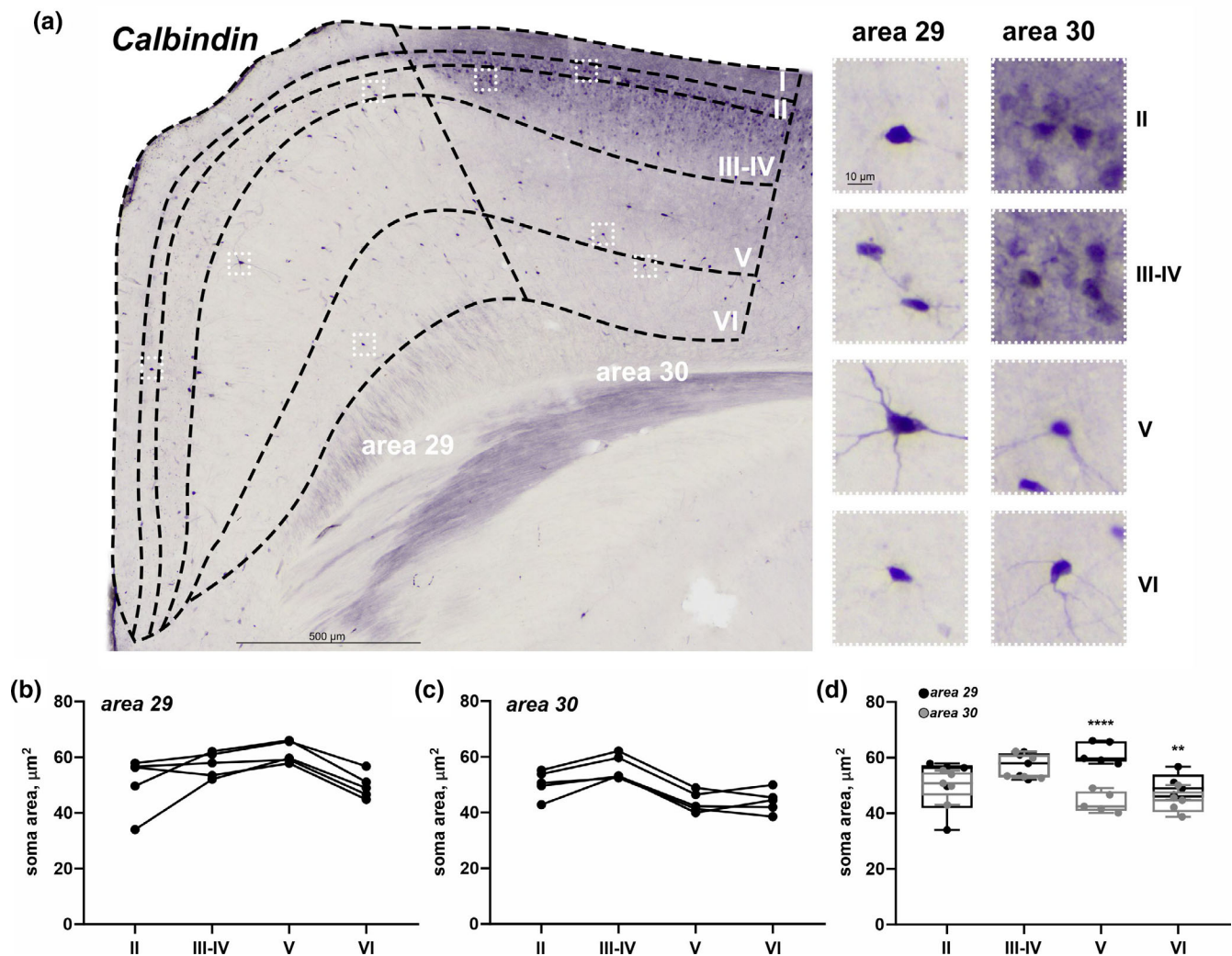


FIGURE 4 Distribution (a) and soma area (b–d) of calbindin-positive neurons in areas 29 and 30 of the cortex of the acorn. Roman numerals—cortical layers. Images within white dotted rectangles are enlarged in the right part of (a). (b and c) Individual data for areas 29 and 30, (d) averaged data. The mean values for animals are shown as individual points. ** $p < 0.01$, **** $p < 0.0001$. Post hoc Bonferroni's test.

significant differences are observed for all layers except for layer I (layer II: $p = 0.0008$, layer III-IV: $p = 0.0013$, layer V: $p = 0.0258$, layer VI: $p = 0.0288$); cellular density in area 30 is higher (Figure 5i).

4 | DISCUSSION

In the present paper, we described the location and division of the retrosplenial cortex in acorn. The border between its two main divisions, the granular and dysgranular areas (areas 29 and 30), can be reliably identified by variations in the expression of NeuN, SMI-32, and CB, as well as differences in neuronal somas, dendritic arborization patterns, and laminar cellular density.

4.1 | Subdivision of the retrosplenial cortex

Domesick (1969), followed by van Groen and Wyss (1990, 1992), subdivided the retrosplenial cortex into a laterally located dysgranular area and a more medially located granular area. The features of this division, wider layers II and III and a thinner layer IV in the dysgranular cortex, as well as differences in soma size, are consistent with those observed in our current study. Based on this, we believe that the lateral and latero-medial regions correspond to the dysgranular area (area 30) and the granular area (area 29), respectively.

The first division of the retrosplenial cortex in rodents was proposed by Brodmann (1909), who distinguished three subareas within area 29: 29a, 29b, and 29c. In the

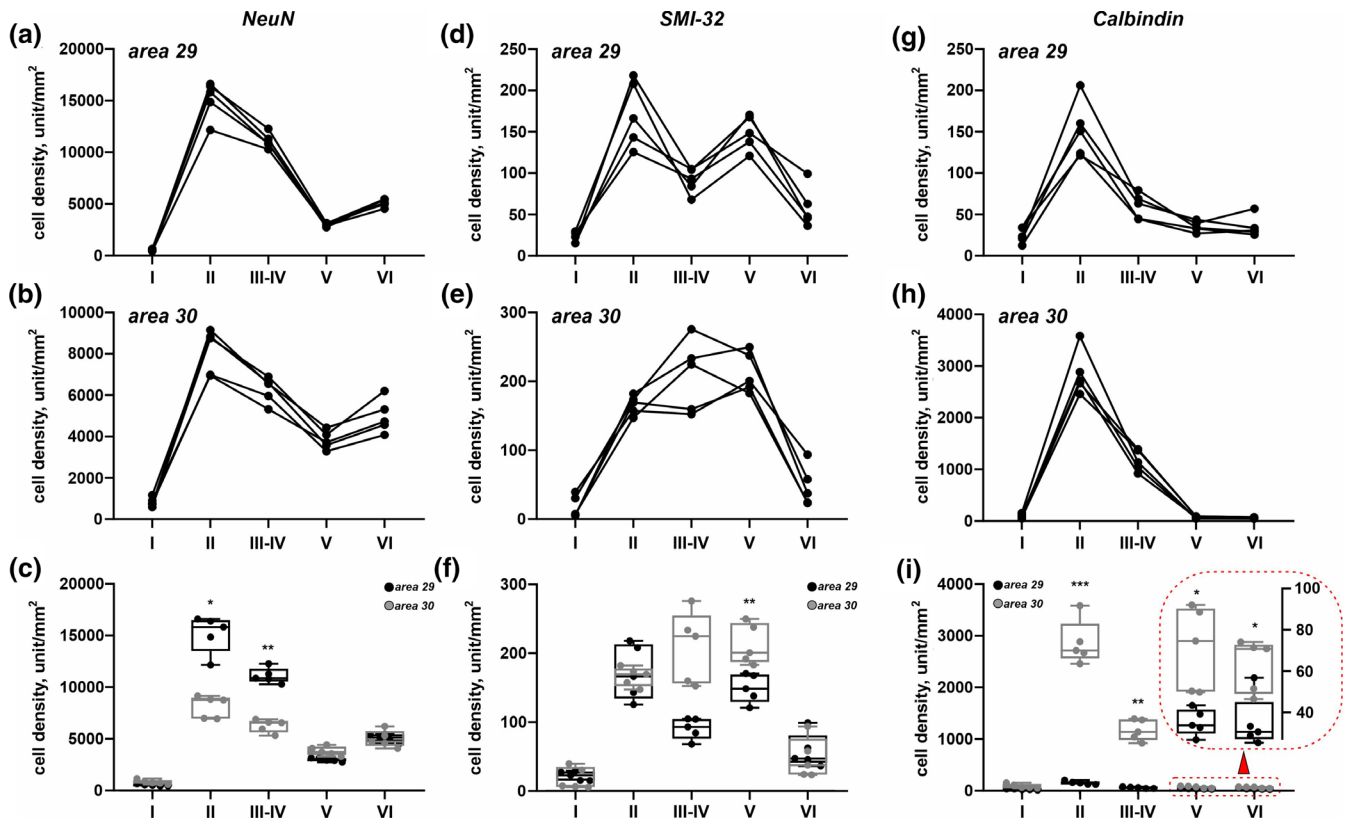


FIGURE 5 Cellular density of NeuN- (a–c), SMI-32- (d–f), and calbindin- (g–i) positive neurons in areas 29 and 30 of the cortex of the acomys. On (c), (f), (i): black—individual data for area 29, gray—individual data for area 30. Roman numerals—cortical layers. The red dotted rectangle in (i) contains an enlarged part of the diagram. The mean values for animals are shown as individual points. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Post hoc Bonferroni's test.

study by Sugar et al. (2011), using NeuN staining, multiple subdivisions of the rat's granular retrosplenial cortex were also identified. Notably, they reported that the ventralmost part of the granular retrosplenial cortex (area 29a) "... differs from the dorsally adjacent 29b since it lacks a fully differentiated layered structure ...". Similar to Vogt and Peters (1981), they noted that, unlike area 29b, which has a thin layer II with densely packed cells, area 29a exhibits a more homogeneous layer II/III. In the acomys, all parts of the granular retrosplenial cortex (area 29) display a prominent, thin layer II with densely packed small cells. van Groen and Wyss (1990) observed in layer II of areas 29a and 29b bands of cells parallel to the pial surface; this feature distinguished these regions from area 29c. The medialmost part of this region was largely devoid of SMI-32+ bundles. It is possible that the border between these two subregions corresponds to the boundary between areas "a" and "b" as described by van Groen and Wyss (1990).

In the mouse, comprehensive axonal connectivity data used as the basis for the Allen Mouse Brain Connectivity Atlas (Allen Institute for Brain Science, 2011) were used to divide the retrosplenial cortex into three areas:

area 29 is non-divided "ventral" granular area, and area 30 which is subdivided into "dorsal" and "agranular" areas (Aggleton et al., 2021). However, in the acomys, applied neuronal markers did not allow us to reveal morphological cues for the division of area 30 into two sub-areas. In conclusion, we find that the main division of the retrosplenial cortex in the acomys is similar to that observed in rats. However, for simplicity and clarity, in this paper we subdivide the retrosplenial cortex only into granular (area 29) and dysgranular (area 30) areas, as other subdivisions are less clearly defined.

4.2 | SMI-32-staining

Using the SMI-32 antibody, Boire et al. (2005) reported that in the hamster cortex, despite the absence of a clear distinction between the retrosplenial and anterior cingulate cortex, increased staining can be observed in the retrosplenial area. Unlike their findings, the present study in the acomys reveals a clear border between the retrosplenial and anterior cingulate cortex, which is evident due to the distinct dendritic bundles (see below). Boire

et al. also noted that the border between the granular and dysgranular areas can be easily visualized because of intense labeling of layers II and III in the granular area, contrasting sharply with very weak labeling in these layers of the dysgranular area. This observation contrasts with our data, as both areas in the *acomys* show intensive SMI-32 labeling in layers II-III.

Similarly, in a study on rat cortex, Kirkcaldie et al. (2002) reported SMI-32+ cells in layer III and layers V and VI for both areas, which is consistent with our findings. In squirrel cortex, Wong and Kaas (2008) described a tri-banded staining pattern with somatic labeling in layers III, V, and VI for the granular area, whereas no pyramidal staining was observed in the dysgranular area; these results are somewhat at odds with our observations. The figures of SMI-32+ labeling in the mouse retrosplenial cortex presented in the article of Van der Gucht et al. (2007) are similar to those obtained in our study: two bands in layers II-III and V and prominent pyramidal cells of layer V in area 30. Additionally, using low-molecular-weight neurofilament protein subunit immunostaining in the mouse cortex, Paulussen et al. (2011) also identified a clear border between the granular and dysgranular retrosplenial cortex.

4.3 | CB-staining

Previously, Frantz and Tobin (1994) documented a “minimal” signal of CB messenger ribonucleic acid in the mouse retrosplenial cortex. Although their main text did not specify the particular area, Figure 2 suggests that the signal was observed in the granular area. Therefore, our findings of extremely weak CB-labeling in area 29 are consistent with data from mice. According to our results, the dysgranular area contains approximately 10 times more CB+ neurons than the granular area. In a comparative analysis by Bjerke et al. (2021), the population of CB+ neurons in mouse brains was examined; they reported a two- to five-fold higher number of labeled cells in the dysgranular area compared to the dorsal and ventral granular areas. This ratio is notably lower than what we observed in the *acomys*, which may reflect more pronounced differences between granular and dysgranular cortical areas in this species.

4.4 | Modular organization of the retrosplenial cortex

A modular organization of the granular retrosplenial cortex in rats was first described by Wyss et al. (1990), who identified apical dendritic bundles organized by pyramidal cells in layer

II. Subsequently, dendritic bundles organized by GABAergic cells in layers II and III were also reported (Ichinohe & Rockland, 2002). These dendritic bundles are believed to be targets of axons originating from the anterior thalamic nuclei (Shibata, 1993; Wyss et al., 1990). In rats, these bundles have also been characterized using various molecular markers, including glutamate receptor subunits 2/3 (GluR2/3) (Ichinohe et al., 2003; Miró-Bernié et al., 2006), GABA receptor type A1 subunit (Miró-Bernié et al., 2006), microtubule-associated protein 2 (Ichinohe et al., 2003), vesicular glutamate transporter 2 (Miró-Bernié et al., 2006), olfactory cell adhesion molecule (Ichinohe et al., 2003), Ca²⁺-binding protein parvalbumin (Ichinohe & Rockland, 2002), and acetylcholinesterase (Wyss & van Groen, 1992a). In our current study, we observed similar dendritic bundles using SMI-32 staining. The width of these bundles ranged from approximately 30 to 100 µm (Ichinohe & Rockland, 2002; Wyss et al., 1990), with center-to-center distances between 38 and 120 µm, which generally aligns with our data. It is important to note that there are no existing data on SMI-32+ dendritic bundles in the retrosplenial cortex of other rodents. In studies employing SMI-32 antibody to investigate the neocortex of mice (Van der Gucht et al., 2007), rats (Kirkcaldie et al., 2002), and hamsters (Boire et al., 2005), no evidence of dendritic bundles in the retrosplenial cortex has been reported based on figures or main text.

4.5 | General conclusions

In the *acomys* brain, the retrosplenial cortex, like the neighboring hippocampus, is larger than in the mouse brain. Note that the retrosplenial cortex and the hippocampus are tightly interconnected (Aggleton et al., 2021). In this regard, our data from *acomys* is corresponding well that in this animal, the hippocampus is more developed than in rodents such as mice or rats, particularly in comparison to the cortex (Dmitrieva et al., 1987; D'Udine & Alleva, 1983). At the same time, the basic features of the retrosplenial cortex in the *acomys* are quite similar to those observed in other rodents. The inter-areal borders may be unambiguously defined using SMI-32 and CB labeling. Despite the overall higher neuronal density in area 29, both SMI-32+ and CB+ neurons are predominantly found in area 30, highlighting their architectonic differences.

5 | LIMITATIONS

First, a comparison between the *acomys* and mouse brains is only qualitative; no statistical comparison was done.

Second, it is well known that sex significantly influences the anatomy and function of the brain (Gualtierotti et al., 2024; Lafta et al., 2024); but in the present study we did not assess differences between *acomys* females and males.

AUTHOR CONTRIBUTIONS

Natalia Merkulyeva: Conceptualization; investigation; funding acquisition; writing – original draft; methodology; validation; visualization; writing – review and editing; software; formal analysis; project administration; data curation; supervision; resources. **Aleksandr Veshchitskii:** Investigation; methodology; visualization; writing – review and editing. **Aleksandr Mikhalkin:** Investigation; visualization; methodology; writing – review and editing. **Anton Beljajev:** Investigation; methodology; visualization; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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